Francesco Maria Vinciguerra

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Milan, Italy & Lausanne, Switzerland

Work Experience

Visiting Researcher — AI Safety & LLM Reliability University of Warwick *2026*

- Conducting research under the supervision of Prof. Paul Jenkins and Dr. Nicola Branchini on estimating the probability of rare behaviours and failure modes in Large Language Models.
- Developing and implementing Monte Carlo sampling methods, including Sequential Monte Carlo and Transition Path Sampling, to efficiently generate rare LLM outputs.
- Comparing the convergence, computational efficiency, and statistical reliability of these methods for rare-event estimation and out-of-distribution evaluation.

Accenture — Data and AI Intern Milan, Italy *Jun 2025 – Sep 2025*

- Benchmarked and evaluated LLM performance across various tasks, systematically assessing reliability, accuracy, and predictive capabilities.
- Architected multi-agent LLM workflows, specifically focusing on reducing hallucinations and structural errors through model routing and specialized prompt engineering.
- Built and validated AI pipelines integrating RAG systems to ground LLM outputs in verified external literature and reduce parameter fabrication.

AI-refinery — Contributor

- Collaborated with the research team to build systematic evaluation frameworks for LLM-based systems, ensuring production robustness.
- Designed and tested agent orchestration pipelines, analyzing outputs to identify and categorize AI reasoning errors.

Education

EPFL — MSc, Applied Mathematics Lausanne, Switzerland *2025 – Present* Focus: *Stochastic Simulation, Statistical Inference, Mathematical Modeling, Deep Learning.*
Research project supervised by Prof. Martin Hairer.

Bocconi University — BSc, Math. and Computing Sciences for AI Milan, Italy *2022 – 2025* **GPA: 28/30.** Thesis: *Geometric Group Theory*.
Relevant Coursework: *Numerical Methods for ODEs, Probability, Bayesian Statistics, Algorithms.*

Selected Projects

AI-project-SuBERT (Computer Vision & NLP) GitHub *Python, PyTorch, OpenCV, BERT*

- Engineered a synthetic data pipeline in Python to overcome data scarcity, implementing custom noise injection, geometric transformations, and blur augmentations.
- Developed a segmentation and recognition workflow for Sumerian glyphs, optimizing inference for real-time applications.
- Integrated a fine-tuned BERT model for the translation of recognized glyph sequences, gaining practical experience with NLP transformer architectures.

Skills & Awards

Programming: Python
AI & LLMs: LLM Evaluation, Agentic Workflows, RAG.
Awards: IMC Trading Challenge (1st Place).
Languages: English (C1), Italian (Native).